Class 5 - Solutions

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Statistical Learning

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Machine Learning and Econometrics

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Class 5 — Statistical Learning Fundamentals: Exercise Solutions

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1 In-Class Exercise #1

Recall the two different fire alarm algorithms.

Algorithm A: when there is no fire the alarm rings 5% of the time; when there is a fire it rings 99% of the time.

Algorithm B: when there is no fire the alarm rings 2% of the time; when there is a fire it rings 90% of the time.

1.1 Q1

Which of these two algorithms has a lower mean squared error? (Reminder: fires are rare.)

Answer.

Because the outcome is binary (fire = 1, no fire = 0), the mean squared error equals the overall misclassification rate. Define the base rate of fires as p (small, since fires are rare — say $p = 0.01$). The MSE for each algorithm is:

$$\text{MSE}_A = p \times (1 - 0.99) + (1 - p) \times 0.05 = 0.01 \times 0.01 + 0.99 \times 0.05 \approx 0.0496$$

$$\text{MSE}_B = p \times (1 - 0.90) + (1 - p) \times 0.02 = 0.01 \times 0.10 + 0.99 \times 0.02 \approx 0.0208$$

Algorithm B has the lower MSE. Even though it misses more fires (10% miss rate vs 1%), fires are so rare that the false-positive rate dominates the overall error. Algorithm A triggers unnecessary alarms 5% of the time it should be silent — and since "no fire" is the common case, this inflates its total error substantially.

1.2 Q2

When should we try to minimize mean squared error, and when is it appropriate to have a different goal?

Answer.

MSE (or misclassification rate) treats all errors as equally costly. This is appropriate when false positives and false negatives have roughly equal consequences.

A different objective is warranted when the **costs of errors are asymmetric**:

- **Missing a fire is far more costly than a false alarm.** An occupant who dies because the alarm failed to ring is a far worse outcome than an unnecessary evacuation. In this case we should weight false negatives heavily — maximising the true-positive rate (sensitivity) matters more than minimising overall errors. A loss function that heavily penalises missed fires is more appropriate than raw MSE.
- **False alarms are very costly.** In some settings (e.g., a hospital surgery suite where every alarm triggers an expensive response), frequent false positives cause real harm. Then we might instead target a very low false-positive rate and accept some missed events.

More generally, MSE is just one loss function. Any time the real-world consequences of different error types differ — spam detection, medical diagnosis, fraud flagging, loan approvals — the loss function should reflect those costs rather than defaulting to MSE.

2 In-Class Exercise #2

After fitting the true function $f(x) = \cos(1.5\pi x)$ to the 30-observation sample, the MSE is not zero.

2.1 Q3

Why is the MSE of the true function not zero?

Answer.

The data were generated as $y_i = f(x_i) + \varepsilon_i$ where $\varepsilon_i \sim N(0, 0.1^2)$ is random noise. Even when we use the exact true function $f(x) = \cos(1.5\pi x)$, the predictions $\hat{y}_i = f(x_i)$ differ from the observed y_i by ε_i. The MSE of the true function therefore estimates:

$$\text{MSE}_{\text{true}} \approx \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i^2 \xrightarrow{n\to\infty} E[\varepsilon^2] = \text{Var}(\varepsilon)$$

This is the **irreducible error** — the component of prediction error that cannot be eliminated no matter how good our model is. It exists because the data-generating process itself is stochastic. Even a perfect model cannot predict noise.

2.2 Q4

How do you expect the MSE of the true function to change as we increase the sample size N?

Answer.

The MSE of the true function **converges to a fixed value** — it does not systematically fall toward zero as N grows.

By the law of large numbers:

$$\frac{1}{N}\sum_{i=1}^{N} \varepsilon_i^2 \xrightarrow{p} E[\varepsilon^2] = \text{Var}(\varepsilon) = 0.01$$

With a small sample ($N = 30$), the sample MSE is a noisy estimate of this quantity and may be somewhat above or below 0.01 due to sampling variation. As N increases, that sampling variation shrinks — the MSE estimate becomes more precise — but it converges to $\text{Var}(\varepsilon)$, not to zero.

This is the defining property of irreducible error: **more data cannot help**. We can reduce the *reducible* part of MSE by improving our model, but we can never reduce the irreducible part.

3 In-Class Exercise #3

3.1 Q5

What degree of polynomial would you choose to give the best predictions $\hat{y}$ in this example?

Answer.

The best degree is the one that **minimises test (out-of-sample) MSE**, not training MSE.

As we increase the polynomial degree:

- **Training MSE** falls monotonically — a higher-degree polynomial can always fit the training points more closely.
- **Test MSE** follows a U-shaped curve: it first falls as the model captures real structure in the data, then rises once the model starts fitting noise (overfitting).

The optimal degree sits at the **bottom of the test-MSE curve**. In the class demonstration the true function is $\cos(1.5\pi x)$, which is well approximated by a low-degree polynomial (roughly degree 3–5). A degree-1 fit is **under-fit** — high bias; degrees beyond roughly 5–6 are **over-fit** — high variance. The train/test MSE plot reveals the crossover point directly.

Practical rule: use a held-out test set (or cross-validation, introduced in Class 9) to plot test MSE against polynomial degree, then pick the degree at the minimum.

> This is the fundamental **bias–variance tradeoff**: flexibility reduces bias but increases variance. The optimal model balances the two.